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Bound by society: An exploratory study of pet attachment and social constraints in pet bereavement among a Turkish sample

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ABSTRACT

Pet owners often form deep emotional bonds with their animals, and the death of a pet can elicit grief responses similar to human loss. However, pet loss is often considered disenfranchised grief, socially unrecognized and unsupported. Guided by social-cognitive processing and attachment perspectives, this study examines how pet attachment, social constraints, and loss-related variables shape grief intensity. The study included 210 adults who lost a pet within the past year and completed measures of pet bereavement, attachment to pets, and social constraints. Attachment to the pet was the strongest predictor of grief severity, while social constraints also influenced grief, with those facing emotional inhibition experiencing greater distress. Loss-related factors such as euthanasia initially predicted grief, but their effect diminished once attachment and social constraints were considered. These findings emphasize the multifaceted nature of pet loss grief and the need for greater societal recognition and support for the bereaved pet owners.

Pets often serve as friends or family members to their owners, who may refer to them as “baby,” “daughter” or “son” (Adrian et al., 2009; Kemp et al., 2016). Owners deeply value the emotional closeness and support their pets provide, making the loss of a pet a profoundly distressing experience that needs mourning (Zilcha-Mano et al., 2012). This grief is akin to that felt after losing a human loved one (Eckerd et al., 2016), with bereaved pet owners experiencing depression, anxiety, sleep disturbances, loss of appetite, social withdrawal, guilt, emptiness, dreams about the pet, longing, complicated grief, prolonged grief disorder, and a sense that life has lost its meaning (Ahn et al., 2023; Karasu & Alkar, 2020; Kimura et al., 2014).

Beyond the emotional impact itself, certain personal and situational factors have been associated with heightened grief after pet loss. These include being female, living alone, not having children, and recently experiencing a stressful life event (Cowling et al., 2020; Yüksel et al., 2023). However, studies related to these factors are inconsistent. For example, Hunt and Padilla (2006) and Silva et al. (2022) did not observe any gender differences. Grief may also be more intense for those who were the pet’s primary caregiver or who lost their pet unexpectedly (Blakey et al., 2022; Tzivian et al., 2014). Furthermore, grief tends to be

more intense when a pet’s death results from euthanasia or is unexpected, in comparison to natural causes (Behler et al., 2020). However, deciding on euthanasia, particularly with adequate emotional support, may help mitigate grief and provide emotional relief (Silva et al., 2025). Additionally, younger individuals tend to report more intense grief responses compared to older adults (Hunt & Padilla, 2006). On the other hand, Testoni et al. (2019) reported no significant association between living alone and grief intensity. Similarly, Uccheddu et al. (2019) found that housing conditions and the presence of children were not correlated with any dimensions of grief. In conclusion, it should be considered that studies vary in design and sample characteristics. Most are cross-sectional (e.g., Behler et al., 2020; Cowling et al., 2020; Tzivian et al. 2014), limiting causal inference and tracking of grief over time. Sample sizes range from small (e.g., Gosse & Barnes, 1994, $n=207$) to large (e.g., Tzivian et al. 2014, $n>500$), affecting statistical power and generalizability. Differences also exist in cultural contexts (e.g., Türkiye vs. UK), pet types, and recruitment methods. Overall, grief intensity following pet loss has been linked to several demographic and situational factors, including gender, living arrangements, age, caregiving role, and the

circumstances of the pet's death, though study designs and cultural contexts vary.

Among the variables associated with intensified grief, attachment to the pet stands out as a strong contributing factor (Cowling et al., 2020; Rémillard et al., 2017). Bowlby (1980) conceptualized attachment behavior as “any form of behavior that results in a person attaining or retaining proximity to some other differentiated and preferred individual” (p. 39). Four core components of an attachment bond were identified: proximity maintenance (the drive to stay physically close), separation distress (the emotional response to separation), secure base (a sense of safety facilitating exploration), and safe haven (a source of comfort during stress) (Hazan & Zeifman, 1999; Rosenthal & Kobak, 2010). Threats to the attachment bond, such as the death of a loved one—an attachment figure—may elicit strong emotional and behavioral reactions, including protest, anger, despair, loneliness, or withdrawal. Over time, however, individuals often adapt to the loss by sustaining a symbolic bond with the deceased, regaining a sense of security, and gradually reengaging with life (Bowlby, 1980; Mikulincer & Shaver, 2008).

The four core components of attachment can also be observed in human–pet relationships (Kurdek, 2008; Prato-Previde et al., 2006; Zilcha-Mano et al., 2012), though the strength and nature of these bonds can vary widely across individuals and contexts. While many people seek and appreciate emotional closeness with their pets, perceiving them as providers of support, comfort, and relief in times of need, it is important to note that not all human–animal interactions are characterized by such intense attachment. Nonetheless, when attachment to a pet is especially strong, its loss has repeatedly been shown to predict more severe grief reactions (Barnard-Nguyen et al., 2016; Eckerd et al., 2016). Regardless of species, attachment level is closely related to the severity of grief, with stronger bonds accompanied by deeper sorrow, yearning, and emotional pain after a pet's death (Jordan & Vonk, 2024; Wu & Song, 2025). Research using the Pet Bereavement Questionnaire (PBQ) indicates that grief is marked by sadness, anger, and guilt, strongly tied to depression, anxiety, and reduced well-being. Individuals with stronger attachment appear especially vulnerable to sleep problems, intense yearning, guilt, and functional impairment (Yiu et al., 2024).

On the other hand, unlike the loss of a human loved one, the death of a pet often goes unacknowledged, cannot be publicly mourned, and lacks social support (Adams et al., 2000; Cleary et al., 2022;

Redmalm, 2015). Since society often views pet relationships as less significant than human ones, pet owners may face social constraints for their grief and are expected to move on quickly without issues (Adams et al., 2000). The term “social constraints” describes any situation where individuals feel pressured by others to control, limit, or alter their thoughts, actions, or emotions (Lepore et al., 1996).

According to the Social-Cognitive Processing Model (SCPM; Lepore & Helgeson, 1998), adjustment to stressful life events depends largely on individuals' ability to openly process and make sense of their experiences. A central premise of the model is that sharing thoughts and feelings about a stressful experience helps the adjustment process that can be hindered by social constraints. Disclosure not only provides social support and validation of trauma-related emotions but also enables others to offer new perspectives, help make sense of the experience, and suggest effective coping strategies. This exchange can deepen individuals' insight and reduce physiological arousal linked to trauma, easing stress and facilitating adjustment. These adaptive processes can be interrupted when people are faced with societal constraints, limiting or impeding emotional recovery and resulting in maladjustment (Juth et al., 2015; Lepore & Helgeson, 1998).

The assumptions of the SCPM can be illustrated with the findings from Lepore et al. (1996) study on bereaved mothers. In this longitudinal investigation, intrusive thoughts in the early stages of grief predicted different adjustment outcomes depending on the level of social constraints. Specifically, when mothers perceived high social constraints, intrusive thoughts were associated with increased depressive symptoms over time. In contrast, when social constraints were low, intrusive thoughts were linked to decreased depressive symptoms, suggesting that the ability to openly express and cognitively process the loss facilitated more adaptive adjustment. This pattern exemplifies the model's central proposition that the social environment can either hinder or promote the cognitive processing of traumatic experiences. Juth et al. (2015) demonstrated that social constraints experienced after bereavement were significantly associated with symptoms of stress, depression, somatic complaints, and poorer overall health. Additionally, their results showed that the interaction between social constraints and loss-related intrusive thoughts further intensified depressive symptoms and stress. Brown et al. (2023) studied bereavement in 60- to 70-year-olds who lost a companion animal and found that one-third hesitated to share their loss due to fear of others' reactions. They also

reported poorer physical health and more intense grief. These findings highlight that grief is not merely an individual experience but also a social process, underscoring the importance of understanding the role of the social environment in coping with loss.

In the context of pet loss, such constraints can lead individuals to suppress their emotions, resulting in adverse psychological outcomes, including somatization, depression, anxiety, perceived stress, functional impairment, poor physical health, and even suicidal ideation (Habarth et al., 2017; Packman et al., 2014; Spain et al., 2019). When a loss is not acknowledged by society as legitimate or significant, it can give rise to *disenfranchized grief*, wherein the bereaved are denied the right to mourn openly or to receive validation for their grief (Doka, 1999). According to the study by Adams et al. (2000), bereaved pet owners often experience stigma due to their emotional responses to the loss of their pets and believe that society does not view a pet's death as a loss worthy of mourning.

Regarding studies on pet loss in Türkiye remain limited. Among the few studies available, Karasu and Alkar (2020) demonstrated that individuals who have lost a pet experience intense emotional responses, including grief, guilt, anger, regret, and longing. The study also highlighted engagement in various memorial practices, such as visiting the pet's grave. Furthermore, many bereaved pet owners cope by caring for stray animals and exhibit increased sensitivity toward animal rights issues (Karasu & Alkar, 2020). Expressing and sharing emotions with family and friends emerged as a critical emotional necessity. However, recent qualitative data from Yüksel (2022) revealed that people who have lost pets often face reluctance from others to acknowledge their grief, leaving bereaved pet owners feeling misunderstood and disrespected, and encountering remarks such as, "You're making a drama out of this" or "Don't exaggerate, it's just a dog!"

This study aims to identify predictors of grief severity following pet loss, with a particular focus on the roles of social constraints and attachment. Previous research has highlighted various socio-demographic, loss-related, and relational factors influencing grief intensity in the context of pet bereavement (Archer & Winchester, 1994; Brown & Symons, 2016; Cowling et al., 2020; Kimura et al., 2014; Kwong & Bartholomew, 2011; McCutcheon & Fleming, 2002; Stokes et al., 2002). However, the interplay between social constraints and these factors remains underexplored. Addressing this gap, the present study examines how social constraints interact with socio-demographic

variables and attachment styles to affect grief outcomes in a Turkish sample. This focus is particularly important given Türkiye's unique sociocultural context, where emotional bonds are not limited to household pets but extend to stray animals, which are often cared for collectively and considered part of the communal life. Recent national survey data show that approximately 36% of individuals in Türkiye currently own a pet, up from 23% in 2017 (KONDA, 2024). This rising trend, coupled with widespread emotional attachment to both domestic and stray animals, underscores the need to understand pet-related grief within this culturally distinct setting.

Method

Participants

Using convenience sampling, we recruited participants online from the general population through various social media platforms and mailing lists. The final analysis included 210 participants ($M = 37.23$ years, $SD = 12.32$), of whom 181 were women (86.2%), 131 (62.4%) were cat owners, and 64 (30.5%) were dog owners. Table 1 presents detailed demographic characteristics of the participants and their deceased pets.

Materials

Sociodemographic and loss-related data

The personal information form includes questions about participants and their deceased pets. It covers the gender, age, marital status, and educational background of the bereaved owners, as well as details about the type of pet, the cause and time since loss, and whether the death was expected or involved euthanasia.

Grief severity

The Pet Bereavement Questionnaire (PBQ; Hunt & Padilla, 2006) is a self-assessment scale developed to evaluate the reactions relating to pet loss. It is a 16-item 4-point Likert scale. The psychometric properties of the Turkish scale were assessed by Yüksel et al. (2023). The Turkish form comprises 16 items and three subscales, listed as grief (e.g., "My life feels empty without my pet"), anger (e.g., "I feel angry at the veterinarian for not being able to save my pet"), and guilt (e.g., "I feel bad that I didn't do more to save my pet"). However, in the Turkish version of the PBQ, two items originally categorized under the Anger factor – item 4 (e.g., "I have had nightmares about my pet's death") and item 14 (e.g., "Memories of my

Table 1. Demographic characteristics of the participants and deceased pets.

Study Variables		N	%
Gender	Female	181	86.2
	Male	29	13.8
Level of education	Primary school	2	1.0
	Secondary school	2	1.0
	High school	21	10.0
	Bachelor's degree	185	88.1
Relationship status	In a relationship	46	21.9
	Not in a relationship	79	37.6
	Married	85	40.5
Having Children	Yes	68	32.4
	No	142	67.6
Primary Caregiver	Yes	196	93.3
	No	14	6.7
Past Pet Loss	Yes	142	67.6
	No	68	32.4
Species of deceased pet	Cat	131	62.4
	Dog	64	30.5
	Bird	13	6.2
	Other	2	1
Cause of death	Old age	47	22.4
	Illness	88	41.9
	Accident	26	12.4
	Malpractice/	16	7.6
	post-treatment complication		
	Poisoning/attack	11	5.2
	Unknown	18	8.6
Was it an expected death?	Other	4	1.9
	Yes	64	30.5
	No	146	69.5
		Mean	SD
Time passed since the death (month)		6.5	3.9
Age of the participants		37.2	12.3

pet's last moments haunt me") – were reassigned to the Grief factor. Cronbach's alpha for the total scale was reported as .87, and for its subscales Grief, Guilt, and Anger were reported as .85, .81, and .70 respectively. The total score obtained from the scale ranges between 0 to 48 with higher scores indicating a higher rate of grief symptoms. In the current study, Cronbach's alpha level was .86 for the whole scale, .86 for the Grief subscale, .80 for the Guilt subscale, and .56 for the Anger subscale.

Attachment to pets

The Lexington Attachment to Pets Scale (LAPS; Johnson et al., 1992) is a self-assessment scale developed to evaluate the emotional bond between people and their pets. The psychometric properties of the Turkish version of the scale were assessed by Karameşe (2014). It is a 23-item 4-point Likert scale consisting of 3 subscales: general attachment (e.g., "My pet knows when I'm feeling bad"), people substituting (e.g., "My pet means more to me than any of my friends"), and animal rights/animal welfare (e.g., "I believe that pets should have the same rights and

privileges as family members"). The Cronbach's alpha for the overall scale and its subscales, people substituting, animal rights/animal welfare, and general attachment are reported as .91, .78, .70, .86 respectively. The total score obtained from the scale ranges from 0 to 69. Higher scores on the scale indicate a higher level of attachment to the pet. Participants were asked to consider the relationship before the pet's death while filling out the questionnaire. In the current study, Cronbach's alpha was calculated as .91 for the whole scale, .85 for the general attachment subscale, .80 for the people substituting subscale, and .70 for the animal rights/animal welfare subscale. The total score with its subscales is an indicator of emotional attachment to the pet.

Social constraints

The Social Constraints Scale (SCS; Lepore et al., 1996) is a self-assessment measure designed to evaluate the extent to which bereaved individuals perceive social restrictions when expressing their thoughts and emotions. The scale assesses feelings of being unable to openly share one's grief due to concerns about others' reactions, as well as perceptions of insufficient emotional support from close relationships. It is a 10-item 5-point Likert scale. It consists of five items that assess perceptions of social constraints. Each item is asked twice, once referring to the "most important person" (e.g., "When you talked about the loss of _____, how often did you get the impression that this person did not want to hear about it?") and once referring to "other people" (e.g., "How often have you felt the need to hide your feelings about the loss of _____ because others around you seemed annoyed?"). In the present study, participants were asked to consider their deceased pets when filling out the questionnaire. The psychometric properties of the Turkish scale are assessed by authors Aksöz-Efe and Erdur-Baker (2018). The items in the Turkish version are compatible with the original factor structure of the scale. The internal consistency coefficient of the total scale was found to be .72 for the entire scale. Higher scores indicate a higher level of social constraints. In the current study, Cronbach's alpha internal consistency was .92.

Procedure

Ethics committee approval for this study was obtained (B.30.2.XXX.0.05.05-020-191). Participants in the study were recruited via online methods. Recruitment was conducted through a range of professional social media platforms, including LinkedIn, Twitter, and Facebook groups related to relevant fields, as well as through electronic

Table 2. Descriptive information of variables ($N=210$).

Variables	Min.	Max.	Mean	SD
PBQ	3	47	27.54	9.43
LAPS	29	69	59.24	9.60
SCS	10	49	21.15	9.62

Abbreviations: PBQ: Pet Bereavement Questionnaire, LAPS: Lexington Attachment to Pets Scale, SCS: Social Constraints Scale.

mailing lists distributed through community networks. Initially, 233 participants were recruited. Twenty-two participants were excluded for not meeting the inclusion criterion of having lost a pet within the last 12 months. Additionally, one statistical outlier was excluded based on an extreme score on the grief and attachment scale exceeding three standard deviations from the sample mean. There was no missing data. Those who met the inclusion criteria and provided informed consent were invited to complete an online survey form that included the Personal Information Form, the Pet Bereavement Questionnaire, the Lexington Attachment to Pets Scale, and the Social Constraints Scale. Descriptive statistics of the scales are presented in Table 2. Participation in the study was entirely voluntary, and no incentives were provided due to the absence of external funding.

Statistical analysis

Data were analyzed using SPSS version 25.0. An a priori power analysis was conducted using G*Power 3.1 to determine the minimum required sample size for the planned analyses. For our main analysis (a hierarchical regression with 12 predictors), we used an F-test for the overall R^2 (Linear multiple regression: fixed model, R^2 deviation from zero). Assuming a medium effect size ($f^2 = .15$), $\alpha = .05$, and desired power ($1-\beta$) = .80, the required sample size was calculated approximately as $N=146$. Our final sample ($N=210$) exceeded these thresholds, adequate power to detect effects of at least medium magnitude. One statistical outlier was excluded. Correlation analysis examined relationships between grief, attachment, social constraints, age, and time since pet loss. A four-step hierarchical regression identified predictors of grief severity. Hierarchical regression analyses were performed using the enter method, where variables were entered in sequential blocks based on theoretical relevance. Step 1 included individual factors (age, gender, not having children, living alone, primary caregiver role, prior pet loss); Step 2 added loss-related factors (time since loss, unexpectedness, euthanasia, cause of death); Steps 3 and 4 included attachment level and social constraints, respectively. Model comparisons used Adjusted R^2 . The variables living alone and unexpected death were entered into the model but did not appear in the coefficients table, as they showed high overlap with other demographic or

cause-of-death variables. For instance, 91% of participants living alone also reported not having children (30 out of 33 cases), and all accident-related deaths (26 cases) were simultaneously coded as unexpected deaths, while 72% of illness-related deaths (63 out of 88 cases) were also coded as unexpected. These redundancies created local multicollinearity, leading SPSS to omit the redundant coefficients, although the variables were included in the model specification.

Results

To examine the relationships between the study variables, Pearson's correlation analysis was conducted, and the results are presented in Table 3. The analysis revealed significant associations between grief reactions to pet loss and both the level of attachment to pets and the presence of social constraints surrounding grief. Specifically, individuals who reported stronger emotional bonds with their pets, as measured by the LAPS, were more likely to experience heightened levels of grief after their pet's death.

In addition, individuals who perceived greater social constraints—such as discouragement from expressing emotions or invalidation of their grief—also reported more intense grief reactions. Furthermore, a significant negative relationship was found between age and grief intensity, indicating that younger individuals tend to experience more severe grief. However, no significant association emerged between the time elapsed since the pet's death and grief responses, suggesting that grief may persist over time regardless of how long ago the loss occurred.

Table 4 presents the findings of the hierarchical regression analysis, which was conducted to determine the relative contributions of various predictors to grief severity. In the first step, personal variables—gender, age, not having children, living alone, being the primary caregiver, and previous experiences of pet loss—were entered into the model. These factors accounted for 24% of the variance in grief ($R^2 = .24$, $F(6, 203) = 10.60$, $p < .001$), indicating that individual characteristics meaningfully influence the grief response.

Table 3. Pearson's correlation analysis results for variables.

	1	2	3	4	5
1 SCS	—				
2 PBQ	.39**	—			
3 LAPS	.13**	.55**	—		
4 Age	-.16**	-.41**	-.24**	—	
5 Time	-.08	.01	.03	-.00	—

** $p < .01$; $N=210$.

Abbreviations: SCS: Social Constraints Scale, PBQ: Pet Bereavement Questionnaire, LAPS: Lexington Attachment to Pets Scale, Time: Time passed since the death.

Table 4. Hierarchical regression analysis for variables predicting grief severity.

Predictor variables	Step 1	Step 2	Step 3	Step 4
Step 1				
Age	.01	-.01	-.01	-.01
Being women	.20**	.15*	.02	.04
Not having children	.18*	.15*	.16*	.15*
Being primary caregiver	.07	.07	.05	.05
Past pet loss	-.01	-.01	-.03	-.08
Step 2				
Days since loss		-.01	-.03	-.01
Presence of euthanasia		.39**	.28*	.18
CoD - Old Age		.41*	.20	.17
CoD - Illness		.47*	.28	.22
CoD - Poison		.17	.08	.07
CoD - Unknown		.22	.12	.06
Step 3				
Attachment to pet			.47***	.44***
Step 4				
Social constraints				.26***
R ²	.24	.31	.48	.54
ΔR ²	.22	.26	.44	.50
R ² change	.24	.07	.17	.06

*** $p < .001$; ** $p < .01$; * $p < .05$.

Abbreviations: CoD: Cause of Death; Excluded Variables: Living Alone, Unexpected Loss, CoD Malpractice, CoD Accident.

Dummy coded variables: Gender (0 = male [reference category], 1 = female), Parental Status (0 = having children [reference category], 1 = not having children), Caregiving Status (0 = secondary caregiver [reference category], 1 = primary caregiver), Past Pet Loss (0 = no [reference category], 1 = yes), Presence of Euthanasia (0 = no [reference category], 1 = yes), CoD (0 = all other causes [reference category], 1 = old age), CoD (0 = all other causes [reference category], 1 = illness), CoD (0 = all other causes [reference category], 1 = poison), CoD (0 = all other causes [reference category], 1 = unknown).

In the second step, loss-related variables—such as time since loss, unexpectedness of the loss, presence of euthanasia, and causes of death—were added. This step contributed an additional 7% to the explained variance ($R^2 = .31$, $\Delta F(9, 194) = 2.23$, $p = .022$), how the loss occurred affects how grief is experienced. In the third step, the level of attachment to the pet was included and added a substantial 17% to the model ($R^2 = .48$, $\Delta F(1, 193) = 62.97$, $p < .001$), highlighting the powerful role that emotional closeness plays in grief severity. Finally, social constraints were introduced in the fourth step, contributing an additional 6% to the explained variance ($R^2 = .54$, $\Delta F(1, 192) = 23.62$, $p < .001$). The final model thus accounted for 54% of the total variance in grief, indicating that personal, situational, emotional, and social variables all play a role in shaping grief experiences.

Examining the individual predictors, female gender initially emerged as a significant predictor of grief ($\beta = .20$, $p < .01$), though its influence declined in the later steps and was no longer significant in the final model ($\beta = .04$, $p > .05$), suggesting that other variables may mediate or overshadow this effect. Similarly, euthanasia was a strong initial predictor ($\beta = .39$, $p < .01$), but its effect weakened as attachment and

social factors were introduced ($\beta = .18$, $p > .05$). The cause of death, particularly death due to old age ($\beta = .41$, $p < .05$) and illness ($\beta = .47$, $p < .05$), also strongly predicted grief at first. However, by the final step, both old age ($\beta = .17$, $p > .05$) and illness ($\beta = .22$, $p > .05$) had lost their statistical significance, indicating possible shared variance with attachment or caregiving roles and later with societal constraints.

Notably, attachment to the pet was the most influential predictor in Step 3 ($\beta = .47$, $p < .001$) and remained highly significant in the final model ($\beta = .44$, $p < .01$), underscoring its central role in the bereavement process. Additionally, social constraints ($\beta = .26$, $p < .001$) became a significant factor in the final step, suggesting that perceived lack of support or emotional invalidation can intensify the grief experience. Interestingly, not having children consistently predicted higher levels of grief across the models ($\beta = .18$ to $\beta = .15$, $p < .05$), perhaps reflecting the unique emotional investment individuals without children place in their companion animals.

Overall, these findings emphasize the importance of both emotional and contextual factors in pet loss grief. Strong attachments and limited social support appear to amplify the emotional burden of loss, highlighting the need to recognize and validate the grief of bereaved pet owners.

Discussion

The present study aimed to investigate the factors influencing grief severity following the loss of a pet. Our results showed that even though personal-related variables and loss-related variables played a significant role, the level of attachment to the pet and social constraints served as strong contributors. These findings highlight the crucial role of both emotional bonds and the social environment in shaping the intensity of grief experienced.

The correlation analysis of the study provides valuable insights into the emotional dynamics of pet bereavement and the factors that influence the intensity of grief. By exploring the connections between grief and attachment to the pet, as well as the role of societal constraints, the research highlights how both emotional bonds and social context shape the grief experience. The results of this study align with previous research, indicating that a stronger emotional bond with a pet led to more intense feelings of grief following its death (McConnell et al., 2011). People with a deep emotional connection to their pets often show more profound reactions (Behler et al., 2020).

They may feel guilt from self-blame for not being able to prevent the pet's death or for making the decision to euthanize (Rémillard et al., 2017; Zasloff, 1996) and may also direct anger toward external factors (e.g., a sudden illness or accident) or themselves for not preventing the loss (Walsh, 2009). These results also indicate that social constraints amplify the emotional intensity of bereavement. Specifically, when societal norms discourage individuals from openly expressing their grief, it can lead to prolonged and more intense grief reactions, such as persistent sadness and depression (Juth et al., 2015). Furthermore, it can lead to excessive rumination and emotional suppression that intensify guilt and confusion (Stroebe & Schut, 1999), and difficulty processing emotions like anger and frustration, which are common but challenging aspects of the grieving process (Klass et al., 1996). The results also revealed a negative association between age and the severity of grief symptoms, suggesting that as individuals grow older, they tend to experience less intense feelings of grief, anger, and guilt following the loss of a pet. This finding aligns with previous studies (Cowling et al., 2020; Kimura et al., 2014; Yüksel et al., 2023), which similarly indicate a decline in emotional distress with advancing age. This may be an indication of the improvement of coping skills and coping styles with age (Folkman et al., 1987). Furthermore, no relationship has been found between the time elapsed since death and grief, which is in line with Eckerd et al. (2016). However, McCutcheon and Fleming (2002) investigated different time since loss ranges (e.g. < 1 month, 1–6 months, > 6 months–1 year, and > 1 year) and found mixed results regarding grief symptoms and time since loss. It seems that grief following pet loss varies widely in both intensity and duration. Some individuals may report distress that subsides within days, while others continue to experience grief for a year or longer. In line with previous research (e.g., Eckerd et al., 2016; Yüksel et al., 2023), the current study employed a 12-month time frame; however, this single time span may have limited our ability to detect potential patterns or correlations between grief severity and the passage of time.

The results of the regression analysis conducted to investigate the main hypothesis of the study provide a much more detailed picture of the grieving process experienced after the loss of a pet. It was found that the variables associated with the bereaved person, such as gender, age, not having children, primary caregiver role, living alone, and experience of losing another companion animal, were predictors of the grief intensity accounting for 24% of the variance.

However, they were not sufficient to explain grief severity. Therefore, variables related to the loss itself, which are generally seen as risk factors when it comes to grief severity (Behler et al., 2020; Lykins et al., 2024; Videla, 2024) such as time since the loss, unexpectedness of the loss, the presence of euthanasia, and various causes of death were added to the model. It has been revealed that these are significant factors in explaining grief severity accounting for an additional 7% of the variance. All these results were consistent with the literature (Behler et al., 2020; Cowling et al., 2020; Gosse & Barnes, 1994; McCutcheon & Fleming, 2002; Stokes et al., 2002; Tzivian et al. 2014; Yüksel et al., 2023). Notably, attachment to the deceased pet accounted for the largest proportion of this variance, aligning with previous research that identifies the strength of attachment as a key predictor of grief intensity (Barnard-Nguyen et al., 2016; Eckerd et al., 2016; Field et al., 2009; Gerwolls & Labott, 1994; Gosse & Barnes, 1994). Furthermore, euthanasia and cause of death (particularly illness and old age) emerged as significant predictors of grief. It should be noted that illness and old age may overlap, as aging is often associated with several illnesses. However, as attachment and social constraints were added to the model, the effects of these loss-related factors diminished. This suggests that while the circumstances of the pet's death do influence grief intensity, the quality of the emotional bond with the pet and the social environment in which the bereaved processes their loss play a more substantial role. Attachment likely amplifies the emotional response (Shear et al., 2007), whereas social constraints may hinder healthy grieving by limiting emotional expression (Lepore & Ituarte, 1999), thereby having a stronger impact on grief outcomes than the cause or nature of the loss itself. These findings highlight the importance of considering psychosocial factors alongside loss characteristics to better understand grief reactions (Neimeyer, 2001; Stroebe & Schut, 1999). As results indicated, experiencing social constraints related to the loss of a pet had a significant impact on the intensity of grief. In parallel, studies have shown that social constraints are linked to more adverse outcomes, including somatization, depression, anxiety, and functional impairment following the loss of a companion animal (Gosse & Barnes, 1994; Habarth et al., 2017). The belief that grief in response to pet loss is not a valid form of bereavement (Testoni et al., 2017) often leads to a lack of understanding within the social support systems of bereaved individuals (Neimeyer & Jordan, 2002). From a broader perspective, research has highlighted the critical role of social

acknowledgement in the grieving process. For example, findings from a cross-sectional study showed that social acknowledgement among individuals bereaved by a loved one's suicide predicted the severity of PTSD (Post-Traumatic Stress Disorder) symptoms and complicated grief, with perceived disapproval from the social environment being associated with higher symptom levels after bereavement (Wagner et al., 2012). Similarly, socially acknowledging a loss has been shown to facilitate bereavement adjustment (Zhou et al., 2023). In the context of pet loss, insufficient social acknowledgement may compound the effects of social constraints, further limiting opportunities to express thoughts and emotions surrounding the loss. Consequently, those unable to access such validation are more likely to experience intense grief symptoms, including clinical depression and suicidal ideation (Packman et al., 2014).

Our study is especially valuable in the Turkish context, where pet ownership and human-animal bonds are significant. As of 2024, in Türkiye, 36% of people own a pet, with 13% having a cat and 8% having a dog in their household (KONDA, 2024). Additionally, Türkiye has a unique tradition of caring for stray animals in public spaces, with many people forming strong emotional bonds to these animals providing them food, shelter, and medical care. The loss of both owned and stray animals can be deeply distressing for caretakers. However, unlike human loss, pet bereavement is not widely recognized as a significant or valid form of mourning in Turkish culture. This societal perception may further contribute to the grief experienced by bereaved pet owners. Mourning rituals typically reserved for humans are generally not extended to animals, underscoring a cultural distinction between human-animal relationships. Furthermore, mourning the loss of a stray animal under one's care is often seen as inconsequential or unnecessary. This perspective is reflected in the recent heated debate in Türkiye surrounding a newly passed law permitting the collection of stray animals, particularly street dogs, from their habitats. Under the law, these animals face the possibility of either euthanasia or placement in shelters under uncertain conditions for the remainder of their lives. However, 63% of the Turkish population opposes harsh measures such as the euthanasia of stray animals, while 22% believe that stray animals do not pose any problem (KONDA, 2024). In Türkiye, many people are deeply concerned about the fate of these animals, feels that their concerns are being ignored, and mourns the loss of stray dogs taken by government authorities (Associated Press in Ankara, 2024). Thus, even though mourning because of pet

loss is less accepted, a lot of individuals live with pets or stray animals together and form strong bonds with them.

As an important implication, this study highlights the critical need for greater awareness and validation of the grief experienced by pet owners following the loss of a companion animal. Addressing the social constraints that prevent bereaved individuals from expressing their emotions could reduce the risk of severe grief reactions, such as depression, anxiety, and even suicidal ideation. This underscores the importance of enhancing social support systems and providing mental health resources that acknowledge pet loss as a valid and significant source of grief, thereby fostering a more compassionate environment for those affected. To effectively assist bereaved pet owners, mental health professionals must be attuned to the significant emotional consequences of this loss and create a safe environment that promotes the grief process. Moreover, the attitudes of veterinary staff play a significant role in the bereavement process, as they often accompany pet owners throughout this journey. Therefore, educating veterinary professionals about the nuances of pet bereavement is critical.

Besides its strengths and contributions to existing literature, the research presents several limitations. Notably, the study's sample showed an unequal gender distribution, with a predominance of young and highly educated women without children. Considering that women typically exhibit higher response rates than men in web-based surveys (Becker, 2022), the predominance of female participants in the present study's sample is to be expected. Additionally, women are generally more likely to seek social support but may also report higher emotional invalidation in contexts of disenfranchised grief such as pet loss (Adams et al., 2000). Thus, results may limit generalizability due to the demographically homogeneous sample, highlighting the need for future studies to include more balanced gender distribution and representative demographic characteristics to better capture grief experience. It is also important to mention that the deceased pets were mostly cats (62%) and dogs (30%), restricting the generalizability of the findings. Although we used the total scores of the scales, we reported the alpha coefficients of all subscales for transparency. The internal consistency of the anger subscale of PBQ was relatively low ($\alpha = .56$) in our sample. In both the original and the Turkish validation study, the anger subscale consistently demonstrated the lowest internal consistency. Given the differences in sample size and the fact that the anger subscale was measured only with three items in the Turkish adaptation, this

limitation is not unexpected. However, as our analyses relied on the total PBQ score rather than its subscales, this limitation is unlikely to have influenced the main findings. Nevertheless, future research should further assess and refine the anger subscale to improve its psychometric properties in this context. One limitation of the present study is that some predictors (e.g., living alone, unexpected death) overlapped strongly with related variables (e.g., not having children, cause-of-death). For instance, 91% of those living alone also reported not having children, and all accident-related deaths were simultaneously coded as unexpected. Due to this local multicollinearity, SPSS did not provide separate coefficients for these predictors, although they were included in the model. Future studies could address this issue by recruiting larger and more diverse samples, employing alternative coding strategies (e.g., combining highly overlapping categories). Moreover, conceptualizing 'unexpected death' independently from cause-of-death categories may help disentangle overlapping constructs. Furthermore, while statistically significant results were obtained and contextualized within the existing literature, it is suggested that a qualitative approach could provide a more nuanced examination of the social constraints and stigmatization faced by individuals following the loss of a pet, as well as the subsequent effects on the bereavement process. Utilizing semi-structured interviews could facilitate a deeper exploration of participants' thoughts and emotions regarding their pets' deaths and their perceptions of how others have responded to their loss. This qualitative method could also enhance understanding of the attachment dynamics between humans and their pets, offering richer insights into the nature of these bonds through direct participant narratives. Although the present study focused on grief severity as the primary outcome, future research would benefit from including broader indicators of psychological adjustment, such as depression, anxiety, or impairment in daily functioning. This would help us to gain a better understanding of the impact of pet loss on individuals' overall wellbeing and daily functioning.

In conclusion, the present study highlights the complex nature of grief following pet loss, emphasizing the significant role of emotional attachment, social constraints, and loss-related factors in shaping the grieving experience. The findings reveal that a stronger attachment to a pet explained higher grief intensity, while social constraints that inhibit open mourning further exacerbate emotional distress. These results underscore the need for greater societal recognition of pet loss as a legitimate source of grief,

encouraging more supportive environments for bereaved pet owners - reminding us that the depth of love we feel for our pets deserves an equally compassionate space for mourning their loss.

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References

- Adams, C. L., Bonnett, B. N., & Meek, A. H. (2000). Predictors of owner response to companion animal death in 177 clients from 14 practices in Ontario. *Journal of the American Veterinary Medical Association*, 217(9), 1303–1309. <https://doi.org/10.2460/javma.2000.217.1303>
- Adrian, J. A. L., Deliramich, A. N., & Frueh, B. C. (2009). Complicated grief and posttraumatic stress disorder in humans' response to the death of pets/animals. *Bulletin of the Menninger Clinic*, 73(3), 176–187. <https://doi.org/10.1521/bumc.2009.73.3.176>
- Ahn, J., Lee, S. W., Kim, K., Jin, B., & Chung, U. S. (2023). The relationship between childhood trauma experience and complicated grief: The importance of psychological support for individuals coping with pet loss in Korea. *Journal of Korean Medical Science*, 38(37), e305. <https://doi.org/10.3346/jkms.2023.38.e305>
- Aksoz-Efe, İ., & Erdur-Baker, O. (2018). Social constraints among bereaving Turkish adults: Testing the construct and convergent validity and reliability of the Social Constraints Scale (Scs). *Anadolu Journal of Educational Sciences International*, 8(2), 293–320.
- Archer, J., & Winchester, G. (1994). Bereavement following death of a pet. *British Journal of Psychology (London, England: 1953)*, 85 (Pt 2)(2), 259–271. <https://doi.org/10.1111/j.2044-8295.1994.tb02522.x>
- Associated Press in Ankara. (2024). "Turkey approves 'massacre law' to remove millions of stray dogs", The Guardian, 30 July. Retrieved from <https://amp.theguardian.com/world/article/2024/jul/30/turkey-approves-massacre-law-remove-millions-street-dogs> (Accessed: 6 November 2024).
- Barnard-Nguyen, S., Breit, M., Anderson, K. A., & Nielsen, J. (2016). Pet loss and grief: Identifying at-risk pet owners during the euthanasia process. *Anthrozoös*, 29(3), 421–430. <https://doi.org/10.1080/08927936.2016.1181362>
- Becker, R. (2022). Gender and survey participation: An event history analysis of the gender effects of survey participation in a probability-based multi-wave panel study with a sequential mixed-mode design. *Methods, Data, Analyses: A Journal for Quantitative Methods and Survey Methodology (Mda)*, 16(1), 3–32.
- Behler, A. M. C., Green, J. D., & Joy-Gaba, J. (2020). "We lost a member of the family": Predictors of the grief experience surrounding the loss of a pet. *Human-Animal Interaction Bulletin*, 8(3), 54–70. <https://doi.org/10.1079/hai.2020.0017>

- Blakey, J. M. K., Beadle-Brown, J. L., Lobb, J., & Heath, D. A. (2022). Bereavement for companion animals: Intensity, moderating variables, and effects on wellbeing. *Frontiers in Veterinary Science*, 9, 1022282. <https://doi.org/10.3389/fvets.2022.1022282>
- Bowlby, J. (1980). *Attachment and loss*. (3rd ed.) New York.
- Brown, C. A., Wilson, D. M., Carr, E., Gross, D. P., Miciak, M., & Wallace, J. E. (2023). Older adults and companion animal death: A survey of bereavement and disenfranchised grief. *Human-Animal Interactions*, 11(1). <https://doi.org/10.1079/hai.2023.0017>
- Brown, O. K., & Symons, D. K. (2016). "My pet has passed": Relations of adult attachment styles and current feelings of grief and trauma after the event. *Death Studies*, 40(4), 247–255. <https://doi.org/10.1080/07481187.2015.1128499>
- Cleary, M., West, S., Thapa, D. K., Westman, M., Vesik, K., & Kornhaber, R. (2022). Grieving the loss of a pet: A qualitative systematic review. *Death Studies*, 46(9), 2167–2178. <https://doi.org/10.1080/07481187.2021.1901799>
- Cowling, D. M., Isenstein, S. G., & Schneider, M. S. (2020). When the bond breaks: Variables associated with grief following companion animal loss. *Anthrozoös*, 33(6), 693–708. <https://doi.org/10.1080/08927936.2020.1824651>
- Doka, K. J. (1999). Disenfranchised grief. *Bereavement Care*, 18(3), 37–39. <https://doi.org/10.1080/02682629908657467>
- Eckerd, L. M., Barnett, J. E., & Jett-Dias, L. (2016). Grief following pet and human loss: Closeness is key. *Death Studies*, 40(5), 275–282. <https://doi.org/10.1080/07481187.2016.1139014>
- Field, N. P., Orsini, L., Gavish, R., & Packman, W. (2009). Role of attachment in response to pet loss. *Death Studies*, 33(4), 334–355. <https://doi.org/10.1080/07481180802705783>
- Gerwolls, M. K., & Labott, S. M. (1994). Adjustment to the death of a companion animal. *Anthrozoös*, 7(3), 172–187. <https://doi.org/10.2752/089279394787001826>
- Gosse, G. H., & Barnes, M. J. (1994). Human grief resulting from the death of a pet. *Anthrozoös*, 7(2), 103–112. <https://doi.org/10.2752/089279394787001970>
- Habarth, J., Bussolari, C., Gomez, R., Carmack, B. J., Ronen, R., Field, N. P., & Packman, W. (2017). Continuing bonds and psychosocial functioning in a recently bereaved pet loss sample. *Anthrozoös*, 30(4), 651–670. <https://doi.org/10.1080/08927936.2017.1370242>
- Hazan, C., & Zeifman, D. (1999). Pair bonds as attachments: Evaluating the evidence.
- Hunt, M., & Padilla, Y. (2006). Development of the pet bereavement questionnaire. *Anthrozoös*, 19(4), 308–324. <https://doi.org/10.2752/089279306785415493>
- Johnson, T. P., Garrity, T. F., & Stallones, L. (1992). Psychometric evaluation of the Lexington attachment to pets scale (LAPS). *Anthrozoös*, 5(3), 160–175. <https://doi.org/10.2752/089279392787011395>
- Jordan, A., & Vonk, J. (2024). No loss of support if attached: Attachment not pet type predicts grief, loss sharing, and perceived support. *Anthrozoös*, 37(3), 535–552. <https://doi.org/10.1080/08927936.2024.2327175>
- Juth, V., Smyth, J. M., Carey, M. P., & Lepore, S. J. (2015). Social constraints are associated with negative psychological and physical adjustment in bereavement. *Applied Psychology. Health and Well-Being*, 7(2), 129–148. <https://doi.org/10.1111/aphw.12041>
- Karamese, H. (2014). *Lexington evcil hayvanlara bağlanma ölçeği'nin Türkçe'ye uyarlanması, Türkçe formun geçerlik ve güvenilirlik çalışması*. University of Gaziosmanpaşa.
- Karasu, S., & Alkar, Ö. Y. (2020). Evcil hayvanını kaybeden bireylerin yaşadıkları yas süreci ile ilgili bir nitel araştırma. *Veteriner Hekimler Derneği Dergisi*, 91(2), 86–97. <https://doi.org/10.33188/vetheder.646796>
- Kemp, H. R., Jacobs, N., & Stewart, S. (2016). The lived experience of companion-animal loss: A systematic review of qualitative studies. *Anthrozoös*, 29(4), 533–557. <https://doi.org/10.1080/08927936.2016.1228772>
- Kimura, Y., Kawabata, H., & Maezawa, M. (2014). Frequency of neurotic symptoms shortly after the death of a pet. *The Journal of Veterinary Medical Science*, 76(4), 499–502. <https://doi.org/10.1292/jvms.13-0231>
- Klass, D., Silverman, P. R., & Nickman, S. L. (1996). *Continuing bonds: New understandings of grief*. Taylor & Francis.
- KONDA. (2024). Temmuz. *Temmuz 2024 Barometre: Kamuoyu Araştırması* (Rapor No. 155). Retrieved from <https://konda.com.tr/uploads/2407-temmuz-barometre-155-kamuoyu-finalbb8eac8bbc78c40859ea5d3f3767f13484d977198fd951497c4ff075f29c1511.pdf>
- Kurdek, L. A. (2008). Pet dogs as attachment figures. *Journal of Social and Personal Relationships*, 25(2), 247–266. <https://doi.org/10.1177/0265407507087958>
- Kwong, M. J., & Bartholomew, K. (2011). "Not just a dog": An attachment perspective on relationships with assistance dogs. *Attachment & Human Development*, 13(5), 421–436. <https://doi.org/10.1080/14616734.2011.584410>
- Folkman, S., Lazarus, R. S., Pimley, S., & Novacek, J. (1987). Age differences in stress and coping processes. *Psychology and Aging*, 2(2), 171–184. <https://doi.org/10.1037/0882-7974.2.2.171>
- Lepore, S. J., Silver, R. C., Wortman, C. B., & Wayment, H. A. (1996). Social constraints, intrusive thoughts, and depressive symptoms among bereaved mothers. *Journal of Personality and Social Psychology*, 70(2), 271–282. <https://doi.org/10.1037/0022-3514.70.2.271>
- Lepore, S. J., & Helgeson, V. S. (1998). Social constraints, intrusive thoughts, and mental health. *Journal of Social and Clinical Psychology*, 17(1), 89–106. <https://doi.org/10.1521/jscp.1998.17.1.89>
- Lepore, S. J., & Ituarte, P. H. (1999). Optimism about cancer enhances mood by reducing negative social interactions. *Cancer Research, Therapy and Control*, 8(3), 165–174.
- Lykins, A. D., McGreevy, P. D., Bennett, B., Paul, N. K., & Gotsis, N. (2024). Attachment styles, continuing bonds, and grief following companion animal death. *Death Studies*, 48(7), 698–705. <https://doi.org/10.1080/07481187.2023.2265868>
- McConnell, A. R., Brown, C. M., Shoda, T. M., Stayton, L. E., & Martin, C. E. (2011). Friends with benefits: On the positive consequences of pet ownership. *Journal of Personality and Social Psychology*, 101(6), 1239–1252. <https://doi.org/10.1037/a0024506>
- McCutcheon, K. A., & Fleming, S. J. (2002). Grief resulting from euthanasia and natural death of companion animals. *OMEGA - Journal of Death and Dying*, 44(2), 169–188. <https://doi.org/10.2190/5QG0-HVH8-JED0-ML16>

- Mikulincer, M., & Shaver, P. R. (2008). An attachment perspective on bereavement. In M. S. Stroebe, R. O. Hansson, H. Schut, & W. Stroebe (Eds.), *Handbook of Bereavement Research and Practice: Advances in Theory and Intervention* (pp. 87–110).
- Neimeyer, R. A. (2001). *Meaning reconstruction & the experience of loss*. American Psychological Association.
- Neimeyer, R. N., & Jordan, J. R. (2002). Disenfranchisement as empathic failure: Grief therapy and the co-construction of meaning. In K. Doka (Ed.), *Disenfranchised grief* (pp. 95–117). Champaign: Research Press.
- Packman, W., Carmack, B. J., Katz, R., Carlos, F., Field, N. P., & Landers, C. (2014). Online survey as empathic bridging for the disenfranchised grief of pet loss. *Omega*, 69(4), 333–356. <https://doi.org/10.2190/OM.69.4.a>
- Prato-Previde, E., Fallani, G., & Valsecchi, P. (2006). Gender differences in owners interacting with pet dogs: An observational study. *Ethology*, 112(1), 64–73. <https://doi.org/10.1111/j.1439-0310.2006.01123.x>
- Redmalm, D. (2015). Pet grief: When is non-human life grievable? *The Sociological Review*, 63(1), 19–35. <https://doi.org/10.1111/1467-954X.12226>
- Rémillard, L. W., Meehan, M. P., Kelton, D. F., & Coe, J. B. (2017). Exploring the grief experience among callers to a pet loss support hotline. *Anthrozoös*, 30(1), 149–161. <https://doi.org/10.1080/08927936.2017.1270600>
- Rosenthal, N. L., & Kobak, R. (2010). Assessing adolescents' attachment hierarchies: Differences across developmental periods and associations with individual adaptation. *Journal of Research on Adolescence: The Official Journal of the Society for Research on Adolescence*, 20(3), 678–706. <https://doi.org/10.1111/j.1532-7795.2010.00655.x>
- Shear, K., Monk, T., Houck, P., Melhem, N., Frank, E., Reynolds, C., & Sillowash, R. (2007). An attachment-based model of complicated grief including the role of avoidance. *European Archives of Psychiatry and Clinical Neuroscience*, 257(8), 453–461. <https://doi.org/10.1007/s00406-007-0745-z>
- Silva, I., Jóluskin, G., Vilhena, E., & Byrne, A. (2022). Adaptation of the pet bereavement questionnaire for European Portuguese speakers. *International Journal of Environmental Research and Public Health*, 20(1), 534. <https://doi.org/10.3390/ijerph20010534>
- Silva, M. V., Santos, R. R., & Barbosa, M. (2025). Euthanasia and prolonged grief: A cross-sectional study with bereaved pet owners. *Journal of Veterinary Behavior*, 79, 60–67. <https://doi.org/10.1016/j.jveb.2025.04.007>
- Spain, B., O'Dwyer, L., & Moston, S. (2019). Pet loss: Understanding disenfranchised grief, memorial use, and posttraumatic growth. *Anthrozoös*, 32(4), 555–568. <https://doi.org/10.1080/08927936.2019.1621545>
- Stokes, S., Planchon, L., Templer, D., & Keller, J. (2002). Death of a companion cat or dog and human bereavement: Psychosocial variables. *Society & Animals*, 10(1), 93–105. <https://doi.org/10.1163/156853002760030897>
- Stroebe, M., & Schut, H. (1999). The dual process model of coping with bereavement: Rationale and description. *Death Studies*, 23(3), 197–224.
- Testoni, I., De Cataldo, L., Ronconi, L., & Zamperini, A. (2017). Pet loss and representations of death, attachment, depression, and euthanasia. *Anthrozoös*, 30(1), 135–148. <https://doi.org/10.1080/08927936.2017.1270599>
- Tzivian, L., Friger, M., & Kushnir, T. (2014). Grief and bereavement of Israeli dog owners: Exploring short-term phases pre-and post-euthanization. *Death studies*, 38(2), 109–117. <https://doi.org/10.1080/07481187.2012.738764>
- Testoni, I., De Cataldo, L., Ronconi, L., Colombo, E. S., Stefanini, C., Dal Zotto, B., & Zamperini, A. (2019). Pet grief: Tools to assess owners' bereavement and veterinary communication skills. *Animals*, 9(2), 67. <https://doi.org/10.3390/ani9020067>
- Uccheddu, S., De Cataldo, L., Albertini, M., Coren, S., Da Graça Pereira, G., Haverbeke, A., Mills, D. S., Pierantoni, L., Riemer, S., Ronconi, L., Testoni, I., & Pirrone, F. (2019). Pet humanisation and related grief: Development and validation of a structured questionnaire instrument to evaluate grief in people who have lost a companion dog. *Animals*, 9(11), 933. <https://doi.org/10.3390/ani9110933>
- Videla, M. D. (2024). Companion animal grief: factors linked to symptom intensity and coping resources.
- Wagner, B., Keller, V., Knaevelsrud, C., & Maercker, A. (2012). Social acknowledgement as a predictor of post-traumatic stress and complicated grief after witnessing assisted suicide. *The International Journal of Social Psychiatry*, 58(4), 381–385. <https://doi.org/10.1177/0020764011400791>
- Walsh, F. (2009). Human-animal bonds I: The relational significance of companion animals. *Family Relations*, 58(4), 363–374.
- Wu, Y., & Song, J. (2025). The relationship between pet attachment and pet loss grief in Chinese undergraduates: A conditional process model. *Behavioral Sciences*, 15(4), 431. <https://doi.org/10.3390/bs15040431>
- Yiu, W. W., Cheung, H. N., & Wong, P. W. (2024). Adaptation and validation of the Pet Bereavement Questionnaire (PBQ) for Chinese population. *Animals*, 14(19), 2845. <https://doi.org/10.3390/ani14192845>
- Yüksel, Ö. (2022). Evcil hayvan kaybı: Eşsiz Bağlardan Eşsiz Kedere. In E. Keser (Ed.), *Kayıp ve yas psikolojisi* (pp. 459–467). Nobel Akademik Yayıncılık.
- Yüksel, Ö., Apak, B., & Demirci, Ö. (2023). Turkish version of the Pet Bereavement Questionnaire: Validity, reliability and psychometric properties. *Death Studies*, 47(2), 130–138. <https://doi.org/10.1080/07481187.2022.2034073>
- Zasloff, R. L. (1996). The role of companion animals in mental health. *Anthrozoös*, 9(2), 113–119.
- Zhou, N., Wei, Y., Killikelly, C., Xu, X., Stelzer, E. M., Maercker, A., Xi, J., & Smith, K. V. (2023). The relationship between social acknowledgment and prolonged grief symptoms: A multiple mediation effect of beliefs about the goodness and controllability of grief-related emotions. *European Journal of Psychotraumatology*, 14(2), 2220633. <https://doi.org/10.1080/20008066.2023.2220633>
- Zilcha-Mano, S., Mikulincer, M., & Shaver, P. R. (2012). Pets as safe havens and secure bases: The moderating role of pet attachment orientations. *Journal of Research in Personality*, 46(5), 571–580. <https://doi.org/10.1016/j.jrp.2012.06.005>